

Gate E5 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **PASS**. Evidence: 20_EVIDENCE/E5/.

E5 — Falsification revalidation (no compilation)

Repeated in this cycle (new evidence): `e5_v121_numerics.py` → `E5_V121_NUMERICS.json`, `log 10_LOGS/E5_v121_numerics.log`. 21 exact or rigorous checks of the numerics added in v1.21, all **PASS**. Every predeclared negative control was rejected.

- C.2 bilateral cleaning (C2-1...C2-3): $338 \leq vu^2a_3^3$ and $1388 \leq vu^2a_4^3$, with $a_3 \geq a^3/2$ and $a_4 \geq a^6/2$, on a grid of $a \leq 1/10$ that includes the η_0 schedule $a = (9 \cdot 10^{19})^{-1}$. Also $138/2208 = 1/16$, and $3 \cdot 11 = 33$, $6 \cdot 23 = 138$. Negative controls: the constant $\times 17$ fails; $138/2208 = 1/8$ is false.
- C.3 chain (C3-1...C3-16): $\beta_0 \geq 2^{-66}$, $L_T \leq 46$, $M_T \leq 5153$, $8 \cdot 5153 + 1 \leq 59480 \ln 2$, $4a_0 + 5 \leq 2^{18}$, $(12/7)(5153 + 1/8) \leq 12748 \ln 2$, $D_0 \leq 2^{\{a_0 + 146\}}$, $Z = 191424$, $U = Y + 191432$, $2 \cdot 10^{17} \cdot 392 \leq 2^{67}$, $U + 67 \leq 2^{59517}$. Negative controls: $M \leq 5100$, 59470 , 2^{17} , 12740 , $2^{\{a_0 + 144\}}$, $Z = 191432$ and 2^{66} are each rejected. Logarithms are evaluated with `mpmath` at 60 digits, used only in monotone comparisons whose margins are $\geq 10^{-3}$.
- E.1 normalization (E1-1): identities and bounds for $h \in \{1, 2, 3, 7, 50\}$ with exact rationals.
- (E.2d) error $\leq n/9$ (E2d-1). The negative control $n/12$ is rejected.

Reused (not repeated): the 24 predeclared tests T01–T24 of `run_v1.2_r1` (`run_v1.2_r1/20_EVIDENCE/E5/`). E0 shows their files are unchanged. The mathematics they test is unchanged in v1.21 apart from the renamed subscripts. The T25/Certo item stays as audited (feasibility-only certificate, valid).

Detector regression (mandate §3 E5): `20_EVIDENCE/E8/e8_detector_check.py`. The corrected pattern recognises the real captured Lean 4.28 warning (backticks), single, double and curly quotes, `sorryAx`, and a line break inside the phrase. It does not fire on standard axiom lists, on identifiers containing "sorry", or on `sorryish`.

E5 verdict: PASS (finite and numeric scope; no universal claim is established by these tests).

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